

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected, May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-Founder

Oct. 2025 - Present

East Lansing, MI

- Streamlined member onboarding by automating GitHub team access and Slack workspace invitations, reducing setup time from 15 minutes to under 30 seconds.
- Built QR code event check-in system with AWS S3 bucket caching and unique token generation for automated attendance tracking.
- Established open source development workflow with GitHub Actions CI/CD, enforcing 85% code coverage; maintained comprehensive documentation and contributor guidelines to reduce development friction.

MindMosaic

Product Manager

Aug. 2025 – Nov. 2025

East Lansing, MI

- Led team of 6 to deliver a production-ready full-stack note-taking application featuring a flexible chapter/page model and OAuth2.0 authentication via Google and GitHub providers.
- Implemented dual-database architecture (PostgreSQL + Neo4j) with compound B-tree indexes (`user_id`, `created_at`) to reduce note search latency from 850ms to 440ms.
- Designed interactive concept mapping, allowing users to connect chapters and pages with labeled arrows.

PROJECTS

Mira — AI Assistant | *Next.js, Electron, FastAPI, PostgreSQL, WebHook*

May 2025 – Nov. 2025

Distributed audio analysis platform enabling intelligent background processing, and smart task execution

- Built RAG pipeline merging BM25 keyword matching with Sentence-BERT embeddings (384-dim) enabling context-aware automation.
- Implemented thread-safe multi-tenant service registry using LRU eviction and TTL-based resource cleanup to manage lifecycle for 100+ concurrent networks.
- Developed Electron desktop client with optimized IPC architecture to stream real-time audio data to the backend, utilizing webhook notifications to synchronize backend processing results in real-time.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow*

Feb. 2025

Won Interactive Media Track with real-time computer vision game utilizing pose-estimation and gesture-detection

- Designed low-latency gesture pipeline using MediaPipe Pose to map 33-point skeletal landmarks to game actions, achieving 60 FPS with sub-20ms input lag by optimizing the inference loop.
- Created custom vector-based physics engine with AABB collision detection to map 3D real-world movements into 2D game state in real-time.

TECHNICAL SKILLS

Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitHub Actions, PostgreSQL, Neo4j
Backend Frameworks FastAPI, SQLAlchemy, Redis, Alembic, Node.js
Frontend Development React, Next.js, Tailwind CSS, Electron, Vite
Languages Python, C++, Java, JavaScript/TypeScript, SQL, Bash